

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$\Rightarrow x_{n+1} = x_n - \frac{(e^{x_n} - 2)}{e^{x_n}}$$

$$x_{n+1} = \frac{e^{x_n} \{x_n - 1\} + 2}{e^{x_n}}$$

$$\boxed{x_{n+1} = x_n - 1 + 2e^{-x_n}}$$